Code of Conduct:

I S.Ramya(192524368) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

S.Ramya

Signature of the student:

ASSESSMENT

SHORT ANSWER:-

1. A Camera captures an image that appears blurred due to improper sampling. Explain how sampling affects image quality and suggest a method to correct it.

Sampling and image Quality.

Sampling determines the number of pixels used to represent an image.

Low sampling causes blur and loss of details.

Increase the sampling rate (higher resolution) to improve image quality.

Use anti-aliasing filter before sampling.

2. Given a real-world scene captured under low light, describe how the image sensing and acquisition process influences the final digital image.

Image sensing and Acquisition in low light.

low light reduces the amount of light reaching the sensor.

This increases image noise and reduces brightness

A better sensor, longer exposure, or proper illumination improves image quality.

3. An image shows aliasing artifacts. Identify the Cause using sampling and quantization concepts and acquisition position influences the final digital image.

Aliasing Artifacts.

Aliasing occurs due to insufficient sampling.

Fine details appears as jagged edges or false patterns.

Increase the sampling rate and apply an anti-aliasing filter before sampling

4. Differentiate how pixel resolution and intensity resolution affect image representation in a practical application like medical imaging.

Pixel Resolution Vs Intensity resolution

Pixel resolution: Determines image sharpness & details.

Intensity resolution Determines the number of gray levels or Colours.

In medical imaging, both are important for accurate diagnosis.

5. A surveilliance system requires real-time image processing. classify the level of Computer Vision involved and justify your choice.

It uses high-level Computer Vision

The system detects recognizes, and tracks object in real time.

It supports automatic monitoring and decision-making.

6. Explain how image formation models help in improving image quality in application such as autonomous driving.

Image formation Models in Autonomous driving

Image formation models explain how light forms an image.

They help correct lighting, distoration, and Camera perspective.

This improves lane, pedestrian, and ~~obj~~ obstacle detection.

7. A grayscale image is converted into a digital image. Describe the role of quantization in this process and its impact on image detail.

Role of Quantization

Quantization converts continuous intensity values into digital values.

More quantization levels preserve image details.

Fewer levels cause loss of details and banding.

8. Given a noisy-image acquired from a sensor, explain how the acquisition process contributes to noise and how it can be minimized.

Noise in image acquisition

Noise is caused by low light, sensor limitation or electronic interference.

It reduces image quality

Use better sensors, proper lighting, and noise reduction filters.

9. In an application like facial recognition. explain how different levels of Computer Vision are involved from image acquistion to decision making.

Computer Vision in facial recognition

Image acquisition captures the face

preprocessing improves image quality.

Features are extracted and Compared with a database.

The system identifies or Verifies the person.

10. An image Captured with low resolution fails to detect objects clearly. Explain how sampling and image formation affect object Visibility.

Low Resolution and object Visibility.

Low sampling reduces spatial resolation

objects become blurred and difficult to detect

Increasing sampling rate and improving image formation produce clearer images.